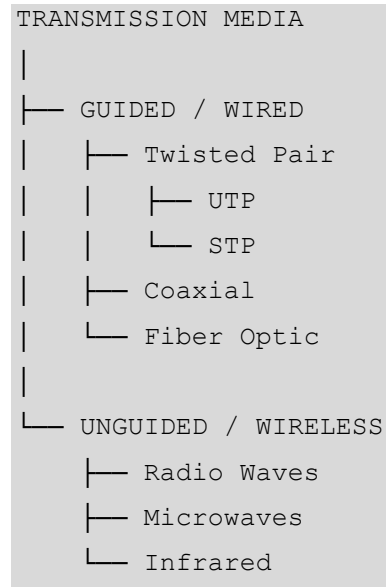


CS331 — Transmission Media Summary Sheet

Transmission media are the **physical paths between the transmitter and receiver**. Your course divides them into **Guided (wired)** and **Unguided (wireless)** media.

Our slides explicitly classify guided media as twisted-pair, coaxial, and fiber-optic, while unguided media transmit signals through the air.



1. Twisted-Pair Cable

- **Category:** Guided / Wired
- **Signal:** Electrical
- **Material:** Two insulated copper wires twisted together
- **Common uses:** LANs and telephone networks

The twisting is important: it **reduces electrical interference**, and according to our slides, **more twists → better quality**.

Type	UTP	STP
Full name	Unshielded Twisted Pair	Shielded Twisted Pair
Shielding	No	Yes
Cost	Cheapest	More expensive
Installation	Easiest	Harder
Weight/thickness	Lighter	Thicker/heavier
EMI protection	Lower	Better
Main advantage	Cheap + easy	Less interference
Main disadvantage	EMI interference	Cost + harder handling

Twisted-Pair Categories

Category	Bandwidth	Maximum Data Rate	Shielding
CAT5e	100 MHz	1 Gbps	UTP or STP
CAT6	250 MHz	1 Gbps	UTP or STP
CAT6a	500 MHz	10 Gbps	UTP or STP
CAT7	600 MHz	10 Gbps	Shielded only
CAT8	2000 MHz	25 or 40 Gbps	Shielded only

A useful pattern:

Higher category → generally higher bandwidth/data capability.

CAT8 can reach **40 Gbps**, but our slides specifically give a distance of **24 m** for shielded CAT8.

Ethernet Wire Labels — Important

This is the TD/RD positive/negative part you mentioned.

Label	Meaning
TD+	Transmit Data — Positive
TD−	Transmit Data — Negative
RD+	Receive Data — Positive
RD−	Receive Data — Negative

So memorize:

T = Transmit **+ = Positive**

R = Receive **- = Negative**

Don't interpret + as "transmission" and - as "receiving." They describe the **positive/negative component of the differential signal**.

For example:

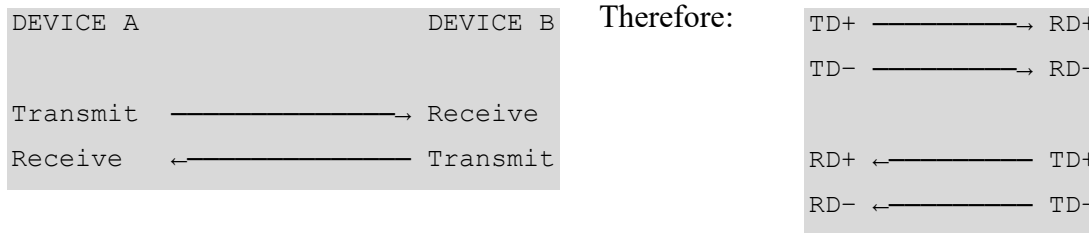
```
TD+ = Transmit, Positive
TD- = Transmit, Negative

RD+ = Receive, Positive
RD- = Receive, Negative
```

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Why crossover cables “cross”

Conceptually, one device’s **transmit** pair needs to reach the other device’s **receive** pair:



That is the easiest way to remember what **crossover** means: **TX crosses to RX**.

2. Coaxial Cable

- **Category:** Guided / Wired
- **Signal:** Electrical
- **Material:** Copper-based cable with shielding.

Think of coaxial as **better protected than ordinary twisted pair**, but bulkier.

Quick memory

Coax = copper + shielding

Its construction helps protect the signal from external interference.

For exam classification:

```
Coaxial
→ Guided
→ Wired
→ Electrical signal
→ Copper
```

3. Fiber-Optic Cable

- **Category:** Guided / Wired
- **Signal:** Light pulses, not electrical signals
- **Material:** Glass/plastic

The cable contains:

```
Core
↓
Cladding
↓
Jacket
```

The **core** carries the light. The **cladding** surrounds the core and has different optical properties.

The **jacket** provides environmental/physical protection.

Pros vs. Cons

Advantages	Disadvantages
Very high bandwidth	Expensive interfaces
Very high data rates	Requires special installation skills
Long distance	More difficult installation
Low attenuation	Higher media/connector cost
Smaller/lighter	
No crosstalk	
Immune to electromagnetic noise	
Highly secure	

Our slides give fiber data rates in the **hundreds of Gbps** and state that it can run **at least 50 km** without repeaters.

Fiber vs. UTP

	UTP	Fiber
Bandwidth	10 Mbps–10 Gbps	10 Mbps–100 Gbps
Distance	1–100 m	1–100,000 m
EMI immunity	Low	Completely immune
Electrical hazards	Low immunity	Completely immune
Cost	Lowest	Highest
Installation skill	Lowest	Highest

So if an exam scenario says:

“We need a long-distance, high-bandwidth connection in an electrically noisy environment.”

→ **Fiber optic**

If it says:

“We need an inexpensive LAN cable that’s easy to install.”

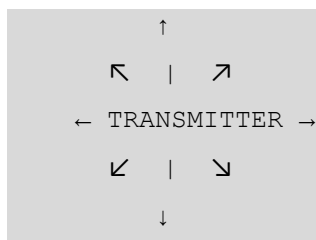
→ **UTP**

4. Radio Waves

Category: Unguided / Wireless

Frequency: < 300 MHz

Direction: Omnidirectional



The signal propagates in **all directions**.

Characteristics

Property	Radio
Direction	Omnidirectional
Frequency	<300 MHz
Distance	Can travel long distances
Typical communication	Broadcast
Examples	AM, FM, TV

When to use

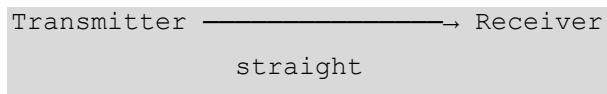
Use radio waves when you want to distribute a signal over a **large area/multiple directions**.

Memory:

RADIO = EVERY DIRECTION

5. Microwaves

- **Category:** Unguided / Wireless
- **Frequency:** 300 MHz–300 GHz
- **Direction:** Unidirectional
- **Propagation:** Line-of-sight



Unlike radio:

Radio = omnidirectional

Microwave = unidirectional

Microwaves travel essentially in straight lines, so transmitter and receiver must be appropriately aligned.

Uses

Our slides give:

Point-to-point links between buildings, wireless networks, satellite communication, cellular phones, and wireless LANs.

Important disadvantage

Very-high-frequency microwaves **cannot penetrate walls well**.

So:

Two buildings need their LANs connected wirelessly with a directional link → **Microwave**

6. Infrared

- **Category:** Unguided / Wireless
- **Frequency:** 300 GHz–400 THz
- **Direction:** Directional
- **Propagation:** Line-of-sight
- **Distance:** Short

Properties

Property	Infrared
Bandwidth	Very large
Range	Short
Wall penetration	Cannot penetrate walls
Propagation	Line-of-sight
Direction	Directional
Outdoors in sunlight	Problematic due to interference

Uses

Our slides list:

Wireless keyboards, wireless mice, remote controls, IrDA communication.

The inability to penetrate walls can actually be useful because the signal remains confined to a room.

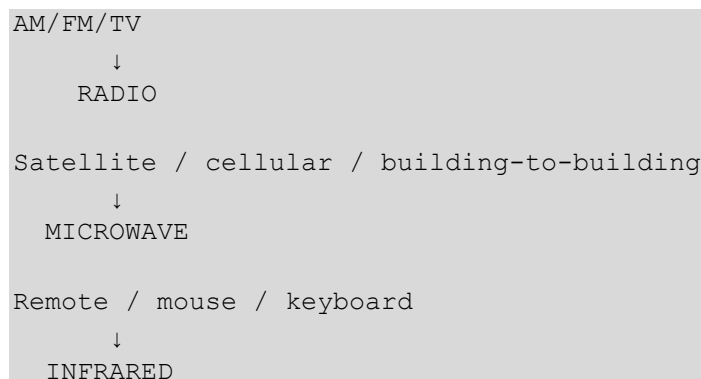
Memory:

INFRARED = REMOTE CONTROL

Wireless Comparison — Memorize This

	Radio	Microwave	Infrared
Frequency	<300 MHz	300 MHz–300 GHz	300 GHz–400 THz
Direction	Omnidirectional	Unidirectional	Directional
Line-of-sight	Not the defining feature	Yes	Yes
Distance	Long	Longer point-to-point	Short
Walls	Better propagation than higher-frequency options	High frequencies struggle	Cannot penetrate
Typical use	AM/FM/TV	Cellular, satellite, WLAN, building links	Remote, mouse, keyboard
Communication style	Broadcast	Unicast / point-to-point	Local point-to-point

The fastest exam distinction is:



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Final “Which Medium?” Cheat Sheet

Scenario	Answer
Cheapest wired LAN	UTP
Need shielding from EMI but still twisted pair	STP
Extremely high bandwidth	Fiber
Very long wired distance	Fiber
Electrical interference is a major problem	Fiber
Highest security among these wired media	Fiber
Simple/cheap installation	UTP
AM/FM/TV	Radio
Need omnidirectional wireless	Radio
Point-to-point between buildings	Microwave
Satellite	Microwave
Cellular	Microwave
Remote control	Infrared
Very short-range line-of-sight	Infrared
Cannot penetrate walls	Infrared / high-frequency microwave
TD+	Transmit Data Positive
TD–	Transmit Data Negative
RD+	Receive Data Positive
RD–	Receive Data Negative

The biggest distinctions I would memorize are **UTP vs STP**, **copper vs light**, **radio = omni**, **microwave = uni/LOS**, **infrared = short/LOS**, and **TD = transmit / RD = receive**. Those give you the answer to most classification and scenario questions from this section.